
Algorithm 1 Cross-Stage Parity Maintenance and Recovery

Require: Ordered protected stages $\mathcal{S}$; parity level $k \in \{1, 2\}$

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1: procedure UPDATEPARITY( $r, q, s, D_s[q]$ )
2:   if  $s \in C[r, q]$  then
3:     return ▷ ignore duplicate contribution
4:   end if
5:   Zero-extend  $P[r, q]$  and  $D_s[q]$  to the longer byte length
6:    $P[r, q] \leftarrow P[r, q] \oplus D_s[q]$ 
7:   if  $k = 2$  then
8:     Zero-extend  $Q[r, q]$  to the same byte length
9:      $Q[r, q] \leftarrow Q[r, q] \oplus g^{\text{rank}(s)} \otimes D_s[q]$ 
10:  end if
11:   $C[r, q] \leftarrow C[r, q] \cup \{s\}$ 
12: end procedure

13: procedure RECOVER( $r, p_f, \mathcal{F}$ )
14:   if  $\mathcal{F} \not\subseteq \mathcal{S}$  or  $|\mathcal{F}| = 0$  or  $|\mathcal{F}| > k$  then
15:     return REPLAYFALLBACK( $r, p_f$ )
16:   end if
17:   Fetch the KV states of all  $s \in \mathcal{S} \setminus \mathcal{F}$ 
18:   if a fetch fails then
19:     return REPLAYFALLBACK( $r, p_f$ )
20:   end if
21:    $N \leftarrow \min_{s \in \mathcal{S} \setminus \mathcal{F}} \text{SLOTS}(D_s)$ 
22:   if  $N = 0$  or the survivor slot-count spread exceeds one then
23:     return REPLAYFALLBACK( $r, p_f$ )
24:   end if
25:   if  $\exists q < N : C[r, q] \neq \mathcal{S}$  or the failed-position input mirror is missing then
26:     return REPLAYFALLBACK( $r, p_f$ )
27:   end if
28:   for all  $x \in \mathcal{F}$  do
29:     ACTIVATEBACKUP( $R(x)$ )
30:   end for
31:   REWIREPIPELINE( $R, \mathcal{F}$ )
32:   if  $|\mathcal{F}| = 1$  then
33:      $x \leftarrow$  the failed stage
34:     for  $q = 0$  to  $N - 1$  do
35:        $\widehat{D}_x[q] \leftarrow P[r, q] \oplus \bigoplus_{s \neq x} \text{PAD}(D_s[q])$ 
36:        $D_x[q] \leftarrow \text{TRIM}(\widehat{D}_x[q], \ell_x)$ 
37:     end for
38:   else
39:      $(x, y) \leftarrow$  the two failed stages ordered by stage rank
40:      $a \leftarrow g^{\text{rank}(x)}; b \leftarrow g^{\text{rank}(y)}$ 
41:     for  $q = 0$  to  $N - 1$  do
42:        $\tilde{P} \leftarrow P[r, q] \oplus \bigoplus_{s \notin \{x, y\}} \text{PAD}(D_s[q])$ 
43:        $\tilde{Q} \leftarrow Q[r, q] \oplus \bigoplus_{s \notin \{x, y\}} g^{\text{rank}(s)} \otimes \text{PAD}(D_s[q])$ 
44:        $\widehat{D}_x[q] \leftarrow (\tilde{Q} \oplus b \otimes \tilde{P}) \oslash (a \oplus b)$ 
45:        $\widehat{D}_y[q] \leftarrow \tilde{P} \oplus \widehat{D}_x[q]$ 
46:        $D_x[q], D_y[q] \leftarrow \text{TRIM}(\widehat{D}_x[q], \ell_x), \text{TRIM}(\widehat{D}_y[q], \ell_y)$ 
47:     end for
48:   end if
49:   for all  $x \in \mathcal{F}$  do
50:     INSTALLKV( $R(x), D_x[0:N]$ )
51:   end for
52:   Execute position  $p_f$  from the mirrored input of the earliest failed stage
53:   return RESUME
54: end procedure
```

Algorithm 2 Recovery-Aware Alternating Layer Placement

Require: Layers $1:L$; devices $\mathcal{D}$; backup hosts $\mathcal{H}$; $I_{\max}$

```
1: function PLACEMENTDP( $S, R, \bar{\psi}$ )
2:    $A[0, 0] \leftarrow (0, 0)$ ; all other states  $\leftarrow (\infty, \infty)$ 
3:   for  $n = 1$  to  $|S|$  do
4:     for  $y = n$  to  $L$  do
5:       for  $l = n - 1$  to  $y - 1$  do
6:          $u \leftarrow (d_n, [l + 1, y])$ 
7:         if  $A[l, n - 1] = (\infty, \infty)$  then
8:           continue
9:         end if
10:        if  $W(u) + B(d_n; R, \bar{\psi}) > M(d_n)$  then
11:          continue
12:        end if
13:         $c \leftarrow T(u)$ 
14:        if throughput mode and  $c > \text{TBT}_{\text{SLO}}$  then
15:          continue
16:        end if
17:         $z \leftarrow (A_{\Sigma}[l, n - 1] + c, \max(A_{\max}[l, n - 1], c))$ 
18:        if  $\rho(z) < \rho(A[y, n])$  then
19:           $A[y, n] \leftarrow z$ ; choice $[y, n] \leftarrow l$ 
20:        end if
21:      end for
22:    end for
23:  end for
24:  return BACKTRACK(choice,  $L, |S|$ )
25: end function

26: procedure ALTERNATINGSEARCH( $\mathcal{D}, \mathcal{H}$ )
27:   best  $\leftarrow \emptyset$ 
28:   for all ordered subsets  $S \subseteq \mathcal{D}$  with  $|S| \geq 2$  do
29:      $\psi \leftarrow \text{INITIALPLACEMENT}(S, L)$ 
30:     if BUILDRECOVERY( $\psi, \mathcal{H}$ ) is infeasible then
31:        $\psi \leftarrow \text{PLACEMENTDP}(S, \emptyset, \psi)$  ▷ repair the seed
32:     end if
33:      $R_{\text{prev}} \leftarrow \perp$ ; best $_S \leftarrow \emptyset$ 
34:     for  $i = 1$  to  $I_{\max}$  do
35:        $R \leftarrow \text{BUILDRECOVERY}(\psi, \mathcal{H})$ 
36:       if  $R$  is infeasible then
37:         break
38:       end if
39:        $\psi' \leftarrow \text{PLACEMENTDP}(S, R, \psi)$ 
40:       if  $\psi'$  is infeasible then
41:         break
42:       end if
43:       consistent  $\leftarrow \text{MEMORYFEASIBLE}(\psi', R, \psi')$ 
44:       if consistent and  $\rho(\psi') < \rho(\text{best}_S)$  then
45:         best $_S \leftarrow (\psi', R)$ 
46:       end if
47:       if consistent and  $\psi' = \psi$  and  $R = R_{\text{prev}}$  then
48:         break
49:       end if
50:        $\psi \leftarrow \psi'$ ;  $R_{\text{prev}} \leftarrow R$ 
51:     end for
52:     if best $_S \neq \emptyset$  and  $\rho(\text{best}_S) < \rho(\text{best})$  then
53:       best  $\leftarrow \text{best}_S$ 
54:     end if
55:   end for
56:   return best
57: end procedure
```
